SECTOR UPDATE

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Samsung Securities

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Rechargeable Batteries (NEUTRAL)

February: Chinese players' market share rising

- The global EV sales volume leapt 25% to 17.79m units in 2024, surging 37% in China and a miniscule 1% in Europe.
- Risks linger for the Korean rechargeable battery value chain, with Chinese battery makers meeting a greater portion of automakers' needs and policy uncertainties still running high.
- In February, we recommend staying conservative towards the sector as: 1) the 4Q earnings season will be wrapped up; and 2) lingering issues persist.

EV Battery Monthly Tracker: 2024 annual figures

Global EV sales volume jumps in 2024: The global EV sales volume leapt 25% to 17.79m units in 2024, according to data published by EV-Volumes—up 37% to 11.5m units in China (65% of the global EV market vs 59% in 2023), up 1% to 3.27m units in Europe, and up 9% to 1.59m units in the US (the slowest annual growth since 2021).

EV battery installations* rise in 2024: EV battery installations grew 24% to 869 GWh in 2024. Chinese cell makers still performed well amid rapid growth of China's EV market. CATL's installations grew 26% to 300 GWh (35% of global total), BYD's rose 43% to 157 GWh (18% of total), and CALB's jumped 49% to 39.8 GWh (beating SK On's and Samsung SDI's).

Among Korea's big-3 cell makers, LG Energy Solution's (LGES) installations rose 4% to 110 GWh, while Samsung SDI's fell 7% to 34 GWh and SK On's slid 8% to 36 GWh. The hike at LGES owed to its shipments to GM (accounting for 16% of its total shipments) soaring 129% thanks to new-car effects, with the increase more than negating the impact of slowing shipments to VW (29% of its total shipments; shipments down 4%) and Tesla (26% of its total; shipments down 15%). Samsung SDI's slump stemmed from its shipments to BMW (37% of its total shipments) falling 5% and to VW (19% of its total shipments) shrinking 19%. SK On's decrease owed to its shipments to HMC (54% of its total shipments) dipping 15% and to VW (12% of its total shipments) sliding 22%.

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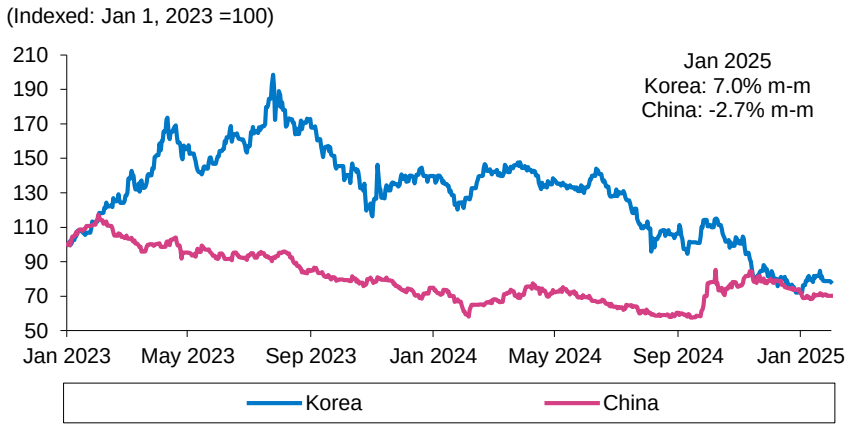
*For passenger autos; estimates derived by multiplying BEV + PHEV sales volume by per-auto battery capacity

Rechargeable battery value chain: Share performance



Note: As of end-Jan 2025
Source: Bloomberg, Samsung Securities

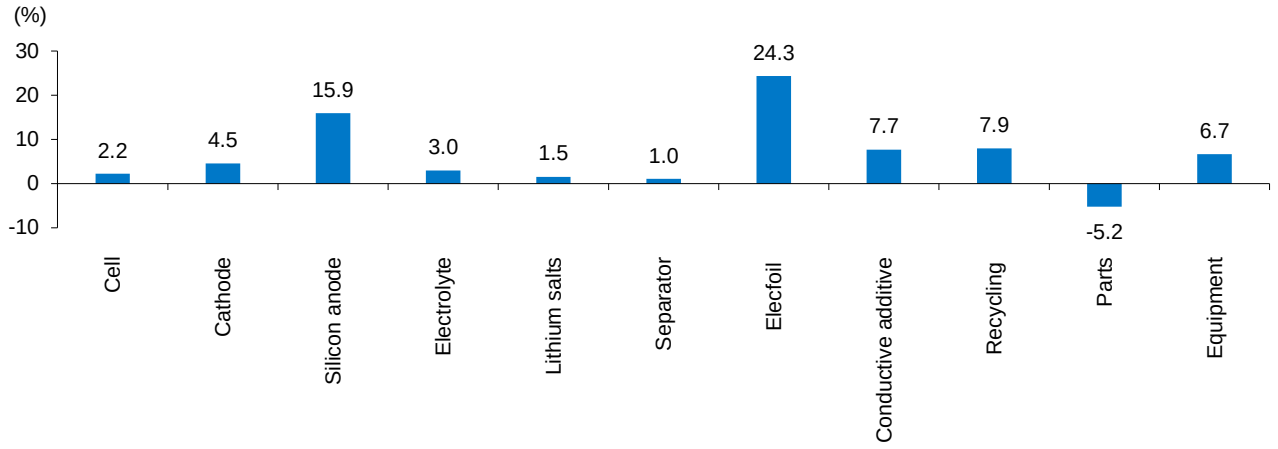
January: Share performance and industry conditions

January—Optimism prevails ahead of Trump's inauguration: In December, the Korean rechargeable battery value chain suffered a severe correction as uncertainties rose after Trump's election victory. In January, shares (especially ones that were oversold) rebounded, backed by aggressive trading ahead of Trump's inauguration—the perception was that negatives had been priced in, though the sector corrected again at end-January on disappointing 4Q results.

Korean battery shares outperform: Shares across the Korean rechargeable battery value chain rose m-m in January (up 7% m-m; climbing m-m for the first time in five months), while the Chinese rechargeable battery value chain lost 2.7% m-m (a car trade-in subsidy in China expired at end-December, causing sluggish EV sales volume in January). Korean cell makers rose 2.2% m-m (Chinese cell makers fell 1.9% m-m) and Korean material makers jumped 10% m-m (Chinese material makers contracted 7.2% m-m).

Elecfoil & silicon anode material segments outperform: Most segments (excluding the parts domain) posted positive share-price returns in January (in December, all segments corrected). The elecfoil segment jumped 24.3% m-m and the silicon anode segment gained 15.9% m-m, the former thanks to SKC shares rallying 43% m-m (on hopes the firm might supply glass substrates to Nvidia, as highlighted during CES; rather than on positives related to elecfoil subsidiary SK Nexilis) and the latter moved up thanks to Daejoo Electronic Materials rising 42% m-m on anticipation that it will supply silicon anodes to Tesla.

Korea's rechargeable battery value chain: Share performance in January, by segment



Source: QuantiWise

Battery installations in 2024—Chinese players' soar while Korean players' plunge: Among Korea's big-3 cell makers, only LG Energy Solution (LGES) saw its installations grow in 2024 (up 4% to 110 GWh). The increase at LGES owed to its shipments to GM (16% of total shipments) soaring 129% on new-car effects (the Equinox EV and Lyriq EV; both feature high-capacity batteries) more than negating the impact of weak shipments to VW (29% of total shipments; shipments down 4%) and Tesla (26%; shipments down 15%). Samsung SDI's installations fell 7%, weighed down by a decline in its shipments to BMW (37% of total shipments; shipments down 5%) and VW (19% of total shipments; shipments down 19%). In contrast, Chinese cell makers recorded installation growth in 2024 backed by solid EV sales volume in China and their meeting a larger portion of global automakers' battery needs. CATL's installations climbed 26%, backed by its shipments to Tesla (15% of total shipments) rising 12% and to BMW (7% of total shipments) growing 22%. BYD's installations rose 43%, backed by a 30% y-y hike in in-house installations (84% of total shipments) and increased shipments to Xiaomi and Nio (thanks to the strong sales volume of their new EVs).

Top-6 battery cell makers: Battery cell installations in 2024, by client

(MWh)		Biggest client	2nd biggest client	3rd biggest client	Other	Total
LGES	Client	VW Group	Tesla	GM		
	Portion (%)	29	26	16	29	100
	Installations	32,079	28,779	17,142	32,033	110,033
	Chg (% y-y)	-4	-15	129		4
Samsung SDI	Client	BMW Group	Rivian Automotive	VW Group		
	Portion (%)	37	19	19	25	100
	Installations	12,614	6,518	6,480	8,412	34,024
	Chg (% y-y)	-5	-2	-19		-7
SK On	Client	Hyundai Motor	Ford	VW Group		
	Portion (%)	54	19	12	15	100
	Installations	19,627	6,876	4,411	5,471	36,386
	Chg (% y-y)	-15	21	-22		-8
CATL	Client	Tesla	Geely Auto Group	BMW Group		
	Portion (%)	15	11	7	66	100
	Installations	46,396	33,373	21,017	199,224	300,009
	Chg (% y-y)	12	35	22		26
BYD	Client	BYD	Xiaomi	Tesla		
	Portion (%)	84	5	3	8	100
	Installations	132,237	7,434	4,783	12,336	156,790
	Chg (% y-y)	30	n/a	85		43
Panasonic	Client	Tesla	Toyota Motor Corp.	Subaru		
	Portion (%)	79	17	2	2	100
	Installations	41,856	9,114	1,267	831	53,068
	Chg (% y-y)	-14	57	26		-7

Source: EV Volumes

Metal prices in January—Lithium prices move sideways, cobalt price plunges: At end-January: 1) lithium carbonate prices stood at CNY73.5/kg (or USD10/kg), rising 1% m-m; and 2) nickel prices stood at USD15/kg, falling 2% m-m. In particular, cobalt prices were USD21.49/kg, contracting 12% m-m to reach the lowest figure since 2016. The plunge owed to a worsening of the supply glut, with: 1) Congo (the world's largest cobalt miner) and Indonesia (second largest) increasing production by 39% and 22% y-y, respectively; and 2) demand for LFP batteries (requiring no cobalt) and high-nickel ternary batteries (requiring less cobalt) rising.

Metal price changes

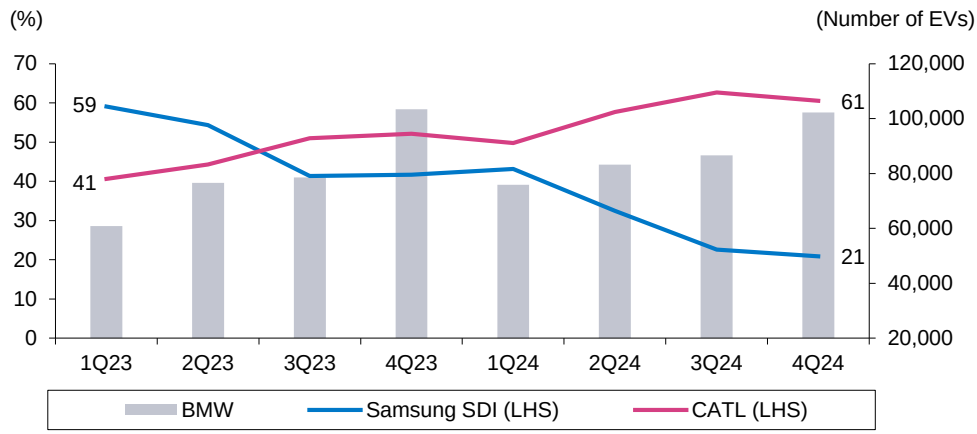
(%)	2021	2022	2023	2024		2025
	Full year	Full year	Full year	4Q	Full year	1Q*
Lithium	445	79	-82	0	-16	1
Nickel	27	45	-46	-10	-6	-2
Cobalt	119	-27	-44	1	-15	-12
Manganese	36	-11	-22	-5	-1	2
Aluminum	42	-16	-1	-1	10	1
Copper	25	-13	1	-11	3	3
Steel	-28	-5	23	7	-28	3

Note: *1Q25 figures are just January
Source: Komis, Samsung Securities

Midstream—Chinese cell makers meeting more of European automakers' battery needs: Looking at battery supply trends in Europe show that Chinese cell makers' shipments to European automakers expanded rapidly in 4Q24, especially their supply of prismatic ternary batteries to BMW (as of end-4Q24, 60% of BMW's EV sales came from Europe; BMW's EV sales in Europe rose 9% in 2024). While CATL's shipments to BMW rose 32% in 2024, Samsung SDI's figure plunged 34%. Accordingly, Samsung SDI met 21% of BMW's battery needs in 4Q24 (vs 59% in 1Q23 and 43% in 1Q24), while CATL met 61% of its battery requirements (vs 41% in 1Q23).

CATL's solid performance owed to: 1) its batteries being used in the BMW 3 Series PHEV and the BMW 5 Series PHEV (replacing Samsung SDI; two key EV models for Samsung SDI in 2023); and 2) healthy sales volume of BMW's BEV models—*eg*, the BMW iX1, iX2, iX3, and the Mini Countryman (Samsung SDI's batteries are not used in these models). Regarding its replacing SDI, CATL possibly offered large price cuts (despite its lacking data to compare supply price) given a worsening of business conditions (*eg*, the cutting of subsidies in key EU nations, such as Germany).

BMW's EV sales volume in Europe vs portions to which Samsung SDI and CATL met BMW's battery needs



Source: EV Volumes

BMW: EV sales volume and battery supplier, by region

(vehicle)	1Q23	2Q23	3Q23	4Q23	1Q24	2Q24	3Q24	4Q24	2024 sales volume (y-y)
BMW EV sales volume - Global	**111,564**	**139,703**	**137,571**	**174,097**	**130,756**	**151,551**	**140,820**	**170,550**	**5%**
EVs with Samsung SDI battery	59,245	69,259	54,525	65,479	49,852	47,692	36,456	39,069	-30%
EVs with CATL battery	51,676	68,249	73,003	97,547	72,019	92,057	89,200	102,258	22%
BMW EV sales volume – Europe	**60,843**	**76,633**	**78,524**	**103,442**	**75,862**	**83,274**	**86,563**	**102,229**	**9%**
EVs with Samsung SDI battery	35,944	41,674	32,447	43,119	32,748	27,045	19,546	21,295	-34%
EVs with CATL battery	24,704	33,936	40,041	53,974	37,760	48,057	54,244	61,886	32%
M/S of Samsung SDI (%)	59	54	41	42	43	32	23	21	
M/S of Samsung CATL (%)	41	44	51	52	50	58	63	61	
BMW EV sales volume – US	**14,699**	**20,170**	**17,279**	**19,126**	**14,131**	**17,856**	**15,900**	**24,000**	**1%**
EVs with Samsung SDI battery	12,991	17,544	12,258	12,704	9,047	11,430	10,481	11,878	-23%
EVs with CATL battery	1,299	1,864	2,734	3,978	3,285	4,418	4,473	4,581	70%

Source: EV Volumes

Midstream—Korea's top-3 battery makers present conservative 2025 sales targets and depend more on AMPC for operating profit: At their 4Q earnings calls, Korea's top-3 cell makers set out their 2025 sales guidance. LGES targets sales growth of 5-10% in 2025 and Samsung SDI targets sales growth at a low-single percentage, with both expecting y-y growth to start in 2Q (after falling y-y in 1Q due to sustained inventory adjustment by clients). Though they foresee a return to sales growth in 2025 (vs sales declines in 2024), they believe their 2025 sales will remain lower than their 2023 figures.

Given the top-3 players' conservative 2025 sales goals, their dependency on AMPC for operating profit should rise further. For instance: 1) LGES is expected to post an operating profit of KRW2.07t in 2025, with AMPCs (from its projected 42 GWh output) reaching KRW2.04t; 2) Samsung SDI is expected to book KRW624b in AMPC in 2025, with AMPCs generating 75% of its total operating profit; and 3) SK On should book KRW646b in AMPCs—excluding this, its operating loss would top KRW1.1t in 2025.

Korean Top-3 battery makers: 2025 guidance

	Sales guidance (y-y)	Operating income (KRWb)	AMPC estimates (KRWb)	AMPC/operating profit (%)	Production in North America (GWh)*
LGES	Up 5-10%	2,077	2,041	98%	42.4
Samsung SDI	Up low-digit percentage**	828	624	75%	12. 3
SK On	n/a	-503	646	nm	10. 5

Note: *Samsung SDI includes Michigan pack production (2.8GWh), **based on EV battery
Source: Company data, Samsung Securities estimates

Policy—Trump's executive orders; lingering tariff risks: On day 1 in office (Jan 20), President Trump signed 'Unleashing American Energy', an executive order dismantling the Biden administration's 'Green New Deal and EV mandate'. One highlight of the order: 'Section 7. Terminating the Green New Deal (a) All agencies shall immediately pause the disbursement of funds appropriated through the Inflation Reduction Act of 2022 (Public Law 117-169) or the Infrastructure Investment and Jobs Act (Public Law 117-58) […] and shall review […] financial disbursements of such appropriated funds for consistency with the law and the policy outlined in section 2 of this order. Within 90 days of the date of this order, all agency heads shall submit a report to the Director of the NEC and Director of OMB that details the findings of this review, including recommendations to enhance their alignment with the policy set forth in section 2'.

Section 2 subsection (e) of the order (regarding the appropriateness of grants issued) establishes that the order is to: '…eliminate the "electric vehicle (EV) mandate" and promote true consumer choice […] by considering the elimination of unfair subsidies and other ill-conceived government-imposed market distortions that favor EVs over other technologies and effectively mandate their purchase…Subsection (e) suggests a possible ban[1]on federal agencies: 1) offering of tax credits to EV buyers (based on the IRA's clean vehicle credit provision (tax code: 30D)); and 2) spending their budget on local battery makers (based on the IRA's advanced manufacturing production credit [AMPC] provision (tax code: 45X))'.

The ban would increase US consumers' EV purchase burden and weigh on earnings across the US battery value chain. As related agency heads' findings of the review should be written by Apr 20 (within 90 days of the date of the order), investors need to remain vigilant and refrain from engaging in aggressive trading.

[1] Rechargeable batteries—Dialogue with the market #28: Trump's executive orders and their implications for Korea's battery value chain

Executive order: Summary and key points from 'Unleashing American Energy' executive order

Sec. 2. Policy (e)	Sec. 7. Terminating Green New Deal
To eliminate the "electric vehicle (EV) mandate" […] by considering the elimination of unfair subsidies and other ill-conceived government-imposed market distortions that favor EVs over other technologies and effectively mandate their purchase by individuals, private businesses, and government entities alike by rendering other types of vehicles unaffordable;	• All agencies shall immediately pause the disbursement of funds appropriated through the Inflation Reduction Act or Infrastructure Investment and Jobs Act • That includes funds for EV charging stations made available through the National Electric Vehicle Infrastructure Formula Program and the Charging and Fueling Infrastructure Discretionary Grant Program • All agencies shall review their policies *etc* for consistency with the law and the policy outlined in section 2 of this order. • Within 90 days of the date of this order, all agency heads shall submit a report to the Director of the NEC and Director of OMB that details the findings of this review • No funds identified in this subsection (a) shall be disbursed by a given agency until the Director of OMB and Assistant to the President for Economic Policy have determined that such disbursements are consistent with any review recommendations they have chosen to adopt.

Source: The White House, Samsung Securities

Meanwhile, on Feb 1, Trump signed executive orders imposing tariffs on almost all imports from China, Canada, and Mexico (*ie*, 25% tariffs on Mexican and Canadian products and an additional 10% tariff on China). The announcement hit all sectors of the Korean stock market hard. On Feb 3 (US time; one day before the planned implementation of the tariffs), Canada and Mexico reached an agreement with the US to delay the tariffs until Mar 4, 2025. Close monitoring is required.

Meanwhile, the new tariff on China became effective on Feb 4, resulting in tariffs on imports from China—*eg*, on batteries and battery parts—rising from 25% (under the Biden administration) to 35%. In response, China said that it would impose additional tariffs of up to 15% on coal, petroleum, agricultural machinery, and automobile imports from the US from Feb 10.

Final Section 301 tariff increases on imports from China

Technology	Prior Section 301 tariff	New Section 301 tariff	New maximum tariff	New Section 301 tariff date	
				2025	2026
Electric vehicle (EVs)	25%	100%	110%	O	
EV lithium-ion batteries	7.5%	25%	35%	O	
Other lithium-ion batteries	7.5%	25%	35%		O
Battery parts (non-lithium-ion batteries)	7.5%	25%	35%	O	
Natural graphite	0%	25%	35%		O

Source: BNEF, Samsung Securities

Policy—US automakers can negate 35% tariff on Chinese-made LFP batteries with 10% LFP cell price cut: The hike in tariffs on Chinese-made EV batteries to 35% will increase US automakers' purchase burden when buying Chinese-made LFP batteries. But excessive concerns are not warranted. The Ford Mach-E[2] features LFP batteries, and assuming it uses LFP packs (with LFP CTP technology), cell costs would be USD6,112/kWh due to the tariff hike (vs USD4,528/kWh before). Compared to the US sourcing of NCM batteries (not subject to tariffs), Ford would find sourcing LFP CTP from Chinese makers 12% more expensive in terms of cell cost (or 7% more expensive based on pack cost). US tariffs on China may change depending on negotiations between the two nations, but even if a US automaker demands the same cell price as that for NCM batteries (USD5,460/kWh), Chinese LFP makers should still be able to afford the 10% cut in cell price given LFP battery oversupply conditions in China.

Analysis of impact of tariff increase on Ford Mach-E (NCM vs LFP CTP)

	VCTPR[1]	Capacity (kWh)[2]	Cell cost (USD/kWh)[3]	Pack cost (USD)	Cell cost after 35% tariff (USD)	Pack cost (USD)[4]
NCM	40%	70	78	5,460	5,460	7,800
LFP	40%	57	63	3,593	4,851	
LFP CTP	66%	72	63	4,528	6,112	8,372
NCM vs LFP						
Margin change per EV sold (USD)			15	932	-652	-572
Margin change per EV sold (%)			-19	-17	12	7

Note: 1) LFP CTP's VCTPR calculated based on the capacity (72 kWh)
2) Battery capacity is based on the specification disclosed for the LFP version of the Ford Mach-E
3) NCM price is applied based on BNEF's estimated 2024 average cell price for NCM622 as a substitute. (cf. NCM811 price = USD93),
4) Assumes a pack/module/cell cost ratio of 1 to 0.8 to 0.7; using a CTP method, the module cost is removed, assuming the same packaging cost
Source: BNEF, Samsung Securities

[2] Rechargeable batteries: Dialogue with market #9: Why the focus on LFP batteries We estimate that the battery switch in the Ford Mach-E SR (from an NCM battery to an LFP battery) has led to EV price cuts of 7%, while not damaging its driving range.

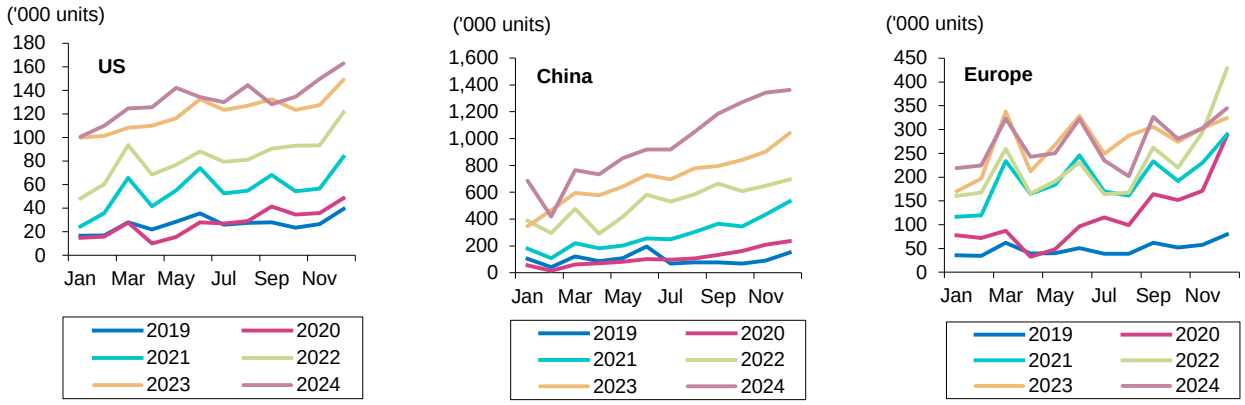
Investment strategy for February—Remain defensive given: 1) ending of 4Q earnings season; and 2) lingering woes: As the 4Q earnings season draws to an end in mid-February, shares should become more volatile as they become more affected by noise. The sector remains mired in uncertainty—*eg*, US tariffs on Mexico and Canada have been postponed until Mar 4; the US's imposition of additional tariffs on China may escalate the US-China tariff war; Germany is set to hold a general election at end-February; and the new European Commission should release new policy measures soon. Investors are advised to remain defensive towards the sector rather than betting big on anticipation that negatives currently weighing on the sector will ease.

Global EV shipments

('000 vehicles)		2023	2024											
	Propulsion	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Americas*	BEV	144.5	92.4	99.2	113.4	123.2	138.1	139.9	132.7	144.6	135.8	139.7	155.0	164.8
	FCEV	0.0	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.1	0.0
	PHEV	40.3	32.4	35.5	40.6	35.9	40.2	33.7	36.0	40.8	34.3	40.8	42.1	49.6
Total		**184.9**	**124.8**	**134.8**	**154.0**	**159.2**	**178.3**	**173.6**	**168.8**	**185.4**	**170.1**	**180.6**	**197.2**	**214.5**
Europe	BEV	233.4	136.5	147.8	217.6	163.7	172.4	233.2	158.3	144.4	238.0	191.5	211.0	241.5
	FCEV	0.1	0.0	0.1	0.4	0.1	0.1	0.0	0.1	0.0	0.1	0.0	0.0	0.0
	PHEV	90.9	82.2	76.5	105.6	78.7	77.6	89.1	77.0	57.6	89.1	88.9	92.1	103.4
Total		**324.4**	**218.7**	**224.4**	**323.6**	**242.5**	**250.1**	**322.4**	**235.3**	**202.0**	**327.1**	**280.5**	**303.2**	**345.0**
China	BEV	683.4	397.1	232.4	459.1	439.1	523.4	520.2	486.3	595.1	692.1	715.0	795.9	786.8
	PHEV	359.1	288.7	185.5	305.0	295.4	329.9	398.9	432.1	453.5	493.8	555.5	546.6	576.7
Total		**1,042.4**	**685.7**	**418.0**	**764.1**	**734.5**	**853.3**	**919.1**	**918.4**	**1,048.6**	**1,186.0**	**1,270.5**	**1,342.5**	**1,363.5**
US	BEV	118.4	76.8	83.1	91.6	99.2	112.1	111.3	105.9	116.4	106.0	107.8	121.9	132.1
	FCEV	0.0	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.1	0.0
	PHEV	30.5	23.8	26.8	33.2	26.5	30.1	22.9	24.0	28.2	22.2	26.7	27.9	30.8
Total		**148.9**	**100.6**	**110.0**	**124.9**	**125.7**	**142.2**	**134.3**	**129.9**	**144.6**	**128.2**	**134.5**	**149.9**	**162.9**
Asia**	BEV	742.6	442.2	275.0	530.0	487.5	578.6	582.5	543.8	653.1	748.6	779.6	873.4	857.4
	FCEV	0.2	0.1	0.4	0.4	0.3	0.4	0.3	0.3	0.4	0.3	0.2	0.1	0.1
	PHEV	367.0	297.7	195.7	317.1	303.9	338.5	409.1	442.8	463.8	505.6	567.0	560.0	589.2
Total		**1,109.8**	**740.0**	**471.2**	**847.4**	**791.7**	**917.5**	**991.9**	**986.9**	**1,117.2**	**1,254.4**	**1,346.8**	**1,433.6**	**1,446.7**
Americas*	BEV	4.1	10.3	8.4	9.1	7.9	9.0	11.1	8.3	8.8	10.9	6.3	7.1	9.9
	PHEV	0.4	1.2	0.8	1.2	1.9	2.4	2.9	2.4	2.3	2.7	2.3	2.2	2.7
Total		**4.6**	**11.5**	**9.2**	**10.3**	**9.8**	**11.4**	**14.0**	**10.7**	**11.1**	**13.6**	**8.6**	**9.3**	**12.6**
Grand total		**1,623.6**	**1,095.0**	**839.6**	**1,335.3**	**1,203.1**	**1,357.3**	**1,501.8**	**1,401.7**	**1,515.8**	**1,765.2**	**1,816.5**	**1,943.2**	**2,018.8**

Note: Passenger cars;
*Includes US;
**Includes China
Source: EV Volumes, Samsung Securities

EV sales volume, by region



Note: Passenger cars, BEV+PHEV
Source: EV-Volumes, Samsung Securities

EV sales volumes: World's top-10 EV makers

Rank	Company	2023	2024											
('000 vehicles)		Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1	BYD	340.3	203.2	126.4	279.2	282.1	294.6	321.9	349.4	397.0	415.6	474.9	445.7	447.5
2	Tesla	195.8	105.6	112.8	167.5	103.5	146.6	193.2	122.5	149.5	191.5	121.3	172.8	202.0
3	GM	93.7	56.1	46.3	53.7	56.5	66.9	56.3	66.9	97.8	110.4	123.6	141.7	144.3
4	Geely Auto Group	71.0	73.2	38.8	52.0	59.7	69.1	75.7	68.8	87.2	103.9	123.6	136.4	127.6
5	VW Group	110.8	69.5	58.0	81.3	76.0	81.1	89.8	74.6	74.9	97.4	100.5	102.0	102.8
6	Chery Automobile	20.5	13.6	15.3	20.5	26.0	33.6	32.0	36.4	40.3	61.8	64.2	71.3	91.5
7	Changan Automobile Group	49.9	50.0	23.9	52.5	42.0	52.1	51.1	48.6	50.6	54.4	68.3	86.9	63.1
8	BMW Group	66.9	41.4	37.3	52.0	47.9	49.2	54.5	51.4	39.9	49.5	51.2	56.7	62.7
9	Li Auto (CHJ Automotive)	50.4	33.0	23.4	32.6	27.9	37.2	49.8	52.8	49.9	55.6	53.3	50.6	60.4
10	GAC	53.5	34.3	18.0	41.7	36.1	42.0	38.4	38.4	39.3	42.2	38.3	43.0	47.6

Note: Passenger cars, BEV+PHEV+FCEV
Source: EV Volumes, Samsung Securities

Battery installations: World's top-10 battery makers

(MWh)	Company	2023	2024											
Rank		Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1	CATL	29,677	20,162	15,305	23,719	21,776	25,572	28,334	24,310	24,512	28,971	27,184	29,666	30,497
2	BYD	12,787	8,324	5,325	11,213	11,005	11,420	12,930	12,872	14,361	16,329	17,482	17,324	18,204
3	LG Energy Solution	8,886	5,218	5,696	9,079	6,866	8,199	10,429	7,814	8,862	11,778	9,782	12,346	13,964
4	CALB	3,916	2,143	1,339	2,852	2,522	3,158	3,224	3,227	3,406	3,705	4,404	4,855	4,948
5	Panasonic	5,416	3,588	3,628	4,359	4,587	4,666	5,230	4,147	4,220	4,844	4,526	4,663	4,612
6	SK On	3,807	2,151	2,230	3,173	2,913	3,337	3,222	3,083	2,987	2,969	3,097	3,429	3,796
7	Samsung SDI	4,206	2,863	2,755	3,509	2,938	2,969	3,059	2,783	2,450	2,404	2,524	2,633	3,138
8	Sunwoda	1,107	578	341	676	756	1,256	1,432	1,461	1,806	2,126	2,256	2,140	2,180
9	SVOLT	642	581	291	552	650	775	963	1,064	1,512	1,870	1,900	2,072	2,121
10	Gotion	1,131	869	465	743	738	788	935	897	1,063	1,455	1,734	2,082	1,917

Source: EV Volumes, Samsung Securities

Capacity: Battery cell and materials makers

('000 tonnes)	Company	2021	2022	2023	2024E	2025E
Cell (GWh)	Samsung SDI	73	87	101	151	196
	LG Energy Solution	170	200	280	335	473
	SK On	40	88	88	118	230
	CATL*	204	318	496	688	880
	Panasonic	49	51	61	61	111
Cathode	LG Chem	70	90	120	140	170
	Ecopro BM	60	125	190	226	334
	L&F	44	100	160	210	210
	Posco Future M	40	40	155	215	345
	Cosmo AM&T	20	20	20	30	60
Anode	Posco Future M	69	82	82	92	132
Copper foil	Lotte Energy Materials	60	60	60	80	111
	Solus Advanced Materials	24	30	48	66	84
	SKC	52	52	96	96	146

Note: *Includes JV capacity
Source: IR, Media, Samsung Securities

Global rechargeable battery value chain: Valuations

	Segment	Company	Mkt cap	Performance (%)		P/E (x)		P/B (x)		EV/EBITDA (x)		ROE (%)		OPM (%)		EPS growth (%)	
			(KRWb)	1m	6m	2024E	2025E	2024E	2025E	2024E	2025E	2024E	2025E	2024E	2025E	2024E	2025E
Korea	Cells	LG Energy Solution*	80,730	-1.0	7.1	n/a	167.6	4.0	3.8	29.5	16.6	-1.8	2.4	6.4	7.2	nm	nm
		Samsung SDI*	14,406	-10.3	-32.3	23.7	14.8	0.7	0.7	11.9	7.8	6.9	4.5	2.2	5.1	-49.3	-6.9
		SK Innovation*	18,547	-1.5	21.6	n/a	25.9	0.6	0.7	17.3	8.7	-6.0	3.3	0.4	2.8	nm	nm
	Cathodes	LG Chem*	15,530	-10.2	-23.1	n/a	19.4	0.5	0.5	9.6	4.9	0.8	2.4	1.9	4.6	-67.7	105.4
		L&F	2,980	-0.1	-10.1	n/a	n/a	3.3	4.1	n/a	55.6	-49.2	-5.0	-4.8	0.4	nm	nm
		Ecopro BM*	11,824	0.6	-32.5	n/a	551.1	9.1	8.8	n/a	47.8	-15.6	0.6	2.3	2.8	nm	nm
		Cosmo AM&T	1,622	-10.7	-55.9	91.5	27.3	3.3	2.9	101.8	n/a	6.8	11.2	4.4	5.0	-21.1	169.1
	Cathodes/ anodes	Posco Future M*	10,086	-13.8	-36.7	n/a	330.0	4.3	4.3	164.7	35.0	-0.9	0.7	0.0	2.5	35.8	-21.7
	Silicon anodes	Daejoo Electronic	1,612	29.6	-7.6	86.9	47.6	9.2	7.6	54.1	30.0	12.4	17.1	3.4	15.7	2960.7	51.9
		Hansol Chemical	1,136	-2.3	-34.1	9.4	8.7	1.2	1.0	9.9	5.8	13.0	12.1	16.1	18.0	10.6	7.0
		Dongjin Semichem	1,316	11.8	-21.4	9.3	n/a	1.4	n/a	6.9	n/a	16.4	n/a	13.4	n/a	n/a	n/a
	Electrolytes	Soulbrain	1,430	5.5	-24.2	9.9	9.2	1.4	1.2	7.1	4.2	15.3	14.0	15.8	20.0	15.8	3.1
		Dongwha Enterprise	428	-5.3	-33.8	n/a	12.3	0.4	0.4	18.3	7.8	-5.5	3.8	-1.7	5.5	nm	nm
		Enchem	2,371	-23.1	-30.8	n/a	n/a	6.8	n/a	104.4	n/a	-18.0	n/a	0.7	n/a	nm	n/a
	Lithium salt	Foosung	524	-3.8	-20.6	n/a	n/a	1.3	n/a	47.6	n/a	-8.6	n/a	-8.8	n/a	nm	n/a
		Chunbo*	389	-0.3	-27.6	n/a	n/a	1.6	1.7	n/a	11.8	-21.8	-5.0	-4.4	4.7	nm	nm
	Separators	SKIET*	1,679	-5.8	-28.6	n/a	n/a	0.7	0.8	34.7	37.5	3.6	-4.2	-133.6	-25.9	nm	nm
		WCP	366	-8.2	-49.3	6.8	49.7	0.4	0.4	17.2	10.3	5.5	0.7	15.2	5.9	nm	nm
	Elecfoil	Lotte Energy Materials*	1,157	3.5	-25.2	47.5	91.1	0.8	0.9	34.1	9.4	0.5	0.6	-7.1	1.7	nm	498.2
		Solus Advanced Materials	674	6.1	-29.0	n/a	n/a	0.9	1.3	154.4	17.2	-3.5	-3.1	-17.0	-0.9	nm	nm
		SKC*	6,029	1.5	31.0	n/a	n/a	4.2	4.7	n/a	31.8	-24.8	-7.0	-13.8	1.8	nm	nm
	Conductive additives	Advanced NanoProducts	771	-2.3	-26.9	99.1	39.1	3.1	2.9	70.1	27.6	3.0	7.8	14.3	12.9	-44.9	97.2
		Jeio	423	-11.3	-14.0	40.5	151.1	2.1	2.0	50.3	43.5	6.2	1.5	10.5	4.4	nm	nm
	Recycling	Sungeel Hitech	490	1.0	-27.2	n/a	n/a	1.9	2.0	n/a	75.7	-12.4	-5.8	-3.4	-4.6	nm	nm
		Sebitchem	97	10.3	-23.9	n/a	n/a	1.5	n/a	n/a	n/a	-1.9	n/a	-10.7	n/a	nm	n/a
	Parts	Sang-A Frontech	237	-20.7	-28.8	41.3	73.7	1.2	1.2	20.4	14.9	3.0	1.7	4.9	3.7	-85.8	84.4
		Shinheung SEC*	176	-10.3	-37.6	9.9	n/a	0.5	n/a	8.2	n/a	5.3	n/a	8.1	n/a	n/a	n/a
		Sangsin EDP	94	-10.0	-38.5	18.8	n/a	0.6	n/a	9.5	n/a	3.5	n/a	10.0	n/a	n/a	n/a
	Equipment	SFA*	675	-2.9	-20.7	n/a	6.0	0.6	0.5	15.4	1.9	-2.0	8.4	4.5	8.9	nm	nm
		Hana Technology	175	-3.1	-21.4	n/a	n/a	1.7	n/a	n/a	n/a	-15.3	n/a	-5.3	n/a	nm	n/a
		People & Technology	945	0.8	-29.2	8.8	6.2	1.7	1.4	10.2	n/a	21.1	26.1	14.1	15.1	42.3	44.4
		Wonik PNE	104	-9.0	-34.0	n/a	n/a	1.2	n/a	n/a	n/a	-43.6	n/a	0.9	n/a	nm	n/a
	Other	Ecopro	8,002	-3.8	-33.6	n/a	n/a	5.2	6.3	n/a	49.9	-12.5	-6.9	4.1	-0.3	nm	nm
		Posco Holdings*	19,582	-9.5	-27.2	16.5	10.5	0.3	0.3	7.3	4.5	2.4	3.2	3.0	4.1	-2.0	3.0
China	Cells	CATL	226,342	5.6	52.9	23.2	17.7	4.8	4.0	n/a	9.0	23.5	23.2	11.4	17.3	25.5	24.8
		BYD	185,203	30.4	53.0	26.4	18.2	5.8	4.3	n/a	6.4	23.8	24.8	5.8	6.4	38.8	27.3
		Gotion High Tech	7,854	9.0	16.5	36.6	28.7	1.6	1.5	n/a	14.6	4.3	5.4	4.6	4.2	-2.1	60.2
		EVE Energy	17,783	2.0	18.3	23.5	15.7	2.4	2.1	n/a	11.8	10.9	13.5	9.1	9.4	18.7	27.9
	Lithium	Tianqi Lithium	10,193	3.9	17.6	n/a	25.4	1.2	1.1	n/a	7.5	-13.5	4.4	82.0	44.2	nm	nm
		Jiangxi Ganfeng	12,998	7.3	32.2	n/a	45.5	1.6	1.5	n/a	24.4	-3.7	2.9	8.3	9.6	-94.3	503.6
	Cobalt	Nanjing Hanrui	2,083	7.5	42.6	59.6	n/a	2.0	n/a	n/a	n/a	3.3	n/a	3.7	n/a	n/a	n/a
		Zhejiang Huayou	10,170	8.5	35.5	15.2	13.1	1.4	1.3	n/a	7.3	9.7	10.2	8.2	8.5	13.5	7.1
		China Molybdenum	30,545	7.2	5.7	13.1	12.0	2.6	2.1	n/a	5.3	24.1	17.7	7.4	12.1	61.2	9.9
	Cathodes/ anodes	Shenzhen Dynanonic	1,917	8.6	37.1	n/a	215.2	1.6	1.6	n/a	14.9	-21.2	-0.7	-11.4	0.0	nm	nm
		Beijing Easpring	4,026	9.3	40.1	22.6	24.5	1.5	1.4	n/a	11.1	7.0	5.8	13.8	8.5	-65.0	33.2
		Ningbo Shanshan	3,265	3.6	-1.8	n/a	n/a	0.7	n/a	n/a	n/a	-1.8	n/a	6.4	n/a	n/a	n/a
	Electrolytes	Guangzhou Tinci	7,417	8.5	32.1	77.9	25.9	2.8	2.6	n/a	15.1	3.6	9.4	15.9	9.6	-59.1	100.3
		Shenzhen Capchem	5,085	-6.0	12.6	27.9	16.5	2.7	2.3	n/a	11.3	9.9	13.8	16.2	16.8	16.0	41.0
	Separators	Cangzhou	1,152	4.5	12.6	34.2	n/a	1.1	n/a	n/a	n/a	3.3	n/a	9.2	n/a	n/a	n/a
		Mingzhu Plastic	2,704	12.6	32.1	53.5	21.1	1.4	1.3	n/a	11.4	2.7	5.9	26.4	15.7	-3.6	19.7

	Segment	Company	Mkt cap	Performance (%)		P/E (x)		P/B (x)		EV/EBITDA (x)		ROE (%)		OPM (%)		EPS growth (%)	
			(KRWb)	1m	6m	2024E	2025E	2024E	2025E	2024E	2025E	2024E	2025E	2024E	2025E	2024E	2025E
Japan	Cells	Panasonic	41,488	13.1	67.1	12.4	13.0	0.8	0.9	5.9	5.6	7.3	7.0	4.2	4.6	52.8	-23.5
		GS Yuasa	2,360	-1.6	4.3	7.5	9.1	0.7	0.7	5.5	4.4	10.3	7.9	7.4	7.7	33.2	13.8
	Cathode	Sumitomo Metal Mining	10,104	2.4	-0.2	14.9	12.7	0.5	0.5	12.4	11.4	3.6	4.3	5.4	4.9	-70.3	56.3
		Tanaka Chemical	156	3.5	-30.1	8.5	n/a	0.9	n/a	6.2	n/a	11.1	n/a	5.8	n/a	n/a	n/a
	Anode	Tokai Carbon	1,862	-1.7	-0.1	19.1	9.6	0.6	0.5	5.1	n/a	3.0	5.0	10.6	8.7	-95.6	1620.1
		Nippon Carbon	481	2.4	-3.3	10.9	10.4	0.9	n/a	5.7	n/a	8.7	n/a	17.4	17.3	1.4	7.1
		Showa Denko	6,643	-6.9	16.7	17.8	12.9	1.1	1.1	8.6	7.0	6.6	9.9	-0.3	6.5	nm	43.5
	Electrolytes	Stella Chemifa	564	-0.9	18.1	22.2	17.2	1.2	1.2	6.1	n/a	5.4	7.0	8.9	12.0	-16.0	61.4
	Separators	Sumitomo Chemical	5,393	1.1	-21.1	n/a	22.7	0.6	0.6	n/a	6.1	-16.4	2.6	-17.6	5.5	nm	nm
		Ashai Kasei	13,359	-5.7	5.5	16.6	12.0	0.7	0.7	6.2	5.9	4.6	6.2	5.1	6.6	nm	54.5
		W-Scope	135	-9.3	-37.7	n/a	n/a	0.3	0.3	n/a	2.3	n/a	-0.1	n/a	11.1	n/a	nm
		Toray Industries	16,626	4.4	48.3	35.2	18.3	1.0	0.9	9.5	8.8	2.8	5.2	2.3	5.4	-27.1	71.6
		Nippon Kodoshi	191	-3.4	-2.5	11.7	10.9	0.8	0.8	8.6	n/a	7.4	7.6	11.6	15.6	-39.9	18.9
	Elecfoil	Furukawa	5,045	2.9	130.3	24.1	21.7	1.6	1.5	7.6	9.9	6.9	7.2	1.1	3.3	-88.2	1,020.6
US	EVs	Tesla	1,688,904	-8.4	80.8	175.3	123.6	16.0	14.8	102.6	61.2	10.5	12.3	7.2	8.9	-43.3	21.5
	Lithium	Albemarle	13,205	-10.6	-5.2	n/a	560.7	1.1	1.0	n/a	10.8	-22.1	3.5	2.6	1.5	nm	nm
		SQM	15,523	0.6	5.5	n/a	11.2	n/a	1.9	9.7	7.2	-6.2	17.4	38.1	30.1	nm	nm
		FMC	6,261	-30.3	-43.5	10.4	9.8	1.0	0.9	12.8	8.0	7.6	9.6	11.9	17.6	-65.2	6.7
Other	Cathodes	Umicore S.A	3,707	1.3	-11.9	n/a	9.8	1.2	1.0	n/a	5.5	-46.4	10.6	3.2	12.3	-29.4	-3.5

Note: *Our coverage; As of Feb 7;
Source: Bloomberg

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